Messaging and Calls — QA Test Cases

Task 4.10 · v1.0 · Owner: Syed Sarim Jawad Zaidi · Based on docs/api/messaging_contract.md and call_contract.md (dev, 20 Oct 2026) and the 4.9 mockups

Purpose: prove that text messages, urgent voice messages and voice / video calls work between a Guardian and a Dependent, that permissions are respected in both directions, and that retries never send twice. Run against Abdullah's backend now (Postman) and against Talha's app when 4.6–4.8 are built. Results go on a copy of the QA Checklist template.

1. Test setup

1.1 Test accounts

Role	Name in tests	Set-up
Dependent	Ammi	Registered, KYC verified, linked to Ali.
Primary Guardian	Ali	Phone + CNIC verified.
Secondary A	Sara	Invited by Ali, verified. Permission communication = ON.
Secondary B	Bilal	Invited by Ali, verified. ALL permissions OFF (communication OFF).
Outsider	Zain	A verified Guardian NOT linked to Ammi (for refusal checks).

1.2 How to run

- **Backend now:** Postman folders 16 Messaging, 17 Voice messages and 18 Calls already pass (2.9 run). The cases below add the app side and the human checks.
- **Backend running:** same as 2.9 ($env:SCHEDULER_ENABLED="true" for the 30 s missed-call job). Missed calls can take up to 1 extra minute.
- **Two phones / emulators:** one logged in as Ali (Guardian app), one as Ammi (Dependent app). A third (Sara) for permission cases.
- **Real audio / video:** needs the real ZegoCloud keys in the backend. Until then the call flow works with a mock join token; mark audio / video quality cases BLOCKED.

2. Test cases

A. Text messages (FR-E1)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
M-01	FR-E1	Ali and Ammi linked.	Ali opens Messages → Ammi → types "Did you eat?" → Send.	201. Bubble on Ali's side; Ammi gets a standard notification titled "Ali" with the text (first 80 characters). Message shows in Ammi's chat (25b).		
M-02	FR-E1	M-01 done.	Ammi replies "Yes" (or taps quick reply "I am fine").	Ali's inbox (17) shows Ammi on top with unread badge 1; Chat tab badge = total_unread.		
M-03	FR-E1	Unread reply from Ammi.	Ali opens the chat. Then reopen it.	POST /messages/read marks 1 read; Ammi sees "Read" under her bubble. Reopening marks 0 (safe to repeat).		

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
M-04	FR-E1	Ali's phone in airplane mode.	Send "I will pick you up" → "Not sent · Tap to retry". Turn network on → tap retry.	Delivered ONCE (server answers 200 already_sent if the first try had arrived). Ammi gets one notification only.		
M-05	FR-E1	Postman as Ali.	Send the same client_id twice with the same text; then the same client_id with a different text.	1st 201, 2nd 200 already_sent = true. Different text: 409 CLIENT_ID_REUSED.		
M-06	FR-E1	—	Send " " (spaces), 2001 characters, exactly 2000 characters, and " hi ".	Spaces only and 2001: 422 VALIDATION_ERROR. 2000: 201. " hi " saved as "hi".		
M-07	FR-E1	Sara and Ali both chat with Ammi.	Sara sends "Salam Ammi". Ali opens his chat with Ammi.	Ali does NOT see Sara's message (each Guardian has a private thread).		
M-08	FR-A4	Bilal has communication OFF.	Bilal tries to open / send to Ammi.	403 MODULE_ACCESS_DENIED. App shows the locked screen (17g); Ammi not in his inbox.		
M-09	FR-A4	Sara chatting with Ammi.	Ali switches Sara's communication OFF. Sara refreshes, tries to send and to read history.	Thread disappears from Sara's inbox; send and history give 403. Ali switches it back ON → history returns.		
M-10	Security	—	Ali → Sara (Guardian to Guardian); Zain → Ammi (not linked).	Both 403 NOT_LINKED_TO_DEPENDENT.		
M-11	FR-E1	35 messages in one chat.	Open the chat, scroll to the top.	First 30 load; older ones load on scroll (next_before); order correct, no duplicates.		

B. Urgent voice messages (FR-E4)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
V-01	FR-E4	Ali in chat with Ammi.	Tap the amber mic (17c), record 10 s, Send.	201, type priority_voice. Ammi's phone gets a CRITICAL voice_message alert even if the app is closed; 25c opens.		
V-02	FR-E4	V-01 received.	Ammi taps Play, then "I heard it".	Audio plays (download uses the login token). Notification acknowledged; Ali sees it was heard.		
V-03	FR-E4	Postman.	GET /messages/{id}/audio without a token; as Zain.	401 without token; Zain gets 404 MESSAGE_NOT_FOUND.		
V-04	FR-E4	Ammi logged in.	Look for a mic in Ammi's chat; send POST /messages/voice as Ammi in Postman.	No mic in the Dependent app; Postman 403 FORBIDDEN.		
V-05	FR-E4	Postman as Ali.	Upload a .txt renamed .m4a; a 3 MB file; duration_s 0 and 121; no audio part.	415 UNSUPPORTED_AUDIO; 413 AUDIO_TOO_LARGE; 422 for the rest.		
V-06	FR-E4	Recorder open.	Keep recording past 2:00.	Recording stops at 2:00 and can be sent; timer shows "of 2:00 max".		
V-07	FR-E4	Same voice message sent twice with one client_id (retry).	Check Ammi's notifications.	One message, one critical alert (200 already_sent on the retry).		

C. Voice and video calls (FR-E2)

ID	Req.	Precondition	Steps	Expected result	Result	Notes / Bug
C-01	FR-E2	Ali and Ammi online.	Ali taps the phone icon (17b) → voice call.	Ali sees "Calling… Ringing" (17e). Ammi gets a critical incoming_call and the full-screen Incoming call (25d).		
C-02	FR-E2	C-01 ringing.	Ammi taps Answer.	Status accepted; both enter the call (17f); audio both ways [needs real ZegoCloud keys].		
C-03	FR-E2	Call connected 1–2 min.	Ali taps End. Then tap End again (or Postman end twice).	Status ended, duration_s ≈ talk time; second end is harmless. Calls tab (17d) shows Ended + duration.		
C-04	FR-E2	Ali calls Ammi.	Ammi taps Decline.	Status declined; Ali's screen leaves "Calling" within ~2 s (polling) and shows a short message.		
C-05	FR-E2	Ali calls Ammi.	Ali hangs up before Ammi answers.	Status cancelled; Ammi gets a standard missed_call.		
C-06	FR-E2	Ali calls Ammi.	Nobody answers. Wait 30 s (+ up to 1 min for the job).	Status missed; Ammi gets missed_call; 17d shows Missed in amber.		
C-07	FR-E2	Ali calls Ammi.	Ali calls her again while it rings; Sara calls Ammi while Ali's call rings.	Ali: same call (200 already_active). Sara: 409 USER_BUSY → app shows "Ammi is on another call".		
C-08	FR-E2	Postman.	Ali tries to accept his own call; accept after the call ended; type "fax".	403 FORBIDDEN; 409 CALL_NOT_RINGING; 422 VALIDATION_ERROR.		
C-09	FR-A4	Bilal (communication OFF).	Bilal calls Ammi; Ammi calls Bilal.	Both 403 MODULE_ACCESS_DENIED; no ring on either phone.		
C-10	FR-E2	Ali and Ammi.	Ali starts a VIDEO call; Ammi answers; switch camera off and on (17f).	Video both ways [real keys]; camera toggle works; End returns to chat.		
C-11	FR-E2	Ammi's app.	Ammi taps "Call" next to Sara in her Messages list (25).	Dependent-to-Guardian call works the same (Sara has communication ON).		

3. ICP-4 live run (about 10 minutes)

Roadmap ICP-4: Guardian–Dependent messaging and calling demoed end-to-end, on two phones.

Step	Who	Action	Everyone should see
1	Ali	Send "Did you eat?" to Ammi.	Ammi's phone shows the message; Ali sees Delivered.
2	Ammi	Reply with quick reply "I am fine".	Ali's inbox badge; opening the chat shows "Read" on Ammi's side.
3	Ali	Send a 10 s urgent voice message.	Ammi's phone alerts loudly; she plays it and taps "I heard it".
4	Ali	Voice call Ammi; Ammi answers; talk 30 s; Ali ends.	Both connected; Calls tab shows Ended + duration.
5	Ammi	Call Ali; Ali does not answer for 30 s.	Missed call shown to Ali.
6	Ali	Turn OFF Sara's communication; Sara tries to message Ammi.	Sara sees the locked screen; Panic still reaches Sara if pressed.

Pass rule: every step behaves as written and no Critical / High bug is open. Save results as Humsafar_QA_ICP4_Results_v1.0.

4. Open points

- Real audio / video quality (C-02, C-10) is BLOCKED until the real ZegoCloud keys are wired into the backend.
- "Ask for access" on the locked screen (17g) has no endpoint yet; skip it unless built.
- Limits (2000 characters, 2 min, 2 MB) and the 30 s ring time are open decisions (sds_clarifications items 76, 79, 81, 84); update the cases if they change.